CSC217 KEY POINTS & PRACTICE QUESTIONS

CSC 217 Course Outline

- **Number Systems:** Binary (0-1), Decimal (0-9), Octal (0-7), and Hexadecimal (0-9, A-F).
- **Logic Gates & Karnaugh Maps (K-maps):** Simplification of logic circuits.
- **Semiconductors:** The physical basis for digital components.
- **Circuit Theory Fundamentals:** Voltage sources, current sources, and resistance.
- **Network Theorems:** Superposition Theorem and Thevenin Theorem.

DIGITAL ELECTRONICS MASTER COMPILATION

PHASE 1: CIRCUIT FUNDAMENTALS & SOURCE THEORY

1. Voltage and Current Sources

- **Voltage Source:** A device that produces a continuous force (electromotive force) to move electrons through a circuit.
  - **Direct Voltage (D.C.):** Polarity remains constant; current moves in one direction (e.g., batteries, cells, D.C. generators).
  - **Alternating Voltage (A.C.):** Polarity reverses periodically (e.g., A.C. generators).
  - **Ideal Voltage Source:** Features **zero internal resistance** ($R_{int} = 0$); output voltage remains constant regardless of load changes.
- **Current Source:** A device providing a regular flow of electrons (current) in a circuit.
  - **Ideal Current Source:** Maintains constant current regardless of the voltage across the load.
- **The Causality Rule:** Voltage is the **cause**; Current is the **effect**. Voltage can exist without current, but current cannot flow without voltage.

2. Resistance, Power, and Ohm's Law

- **Ohm's Law:** Current (I) is directly proportional to Voltage (V) across two points ($V = IR$).
- **Resistance (R):** The property of a conductor that opposes current flow, transforming electrical energy into heat.

2. Resistance, Power, and Ohm's Law

- **Ohm's Law:** Current (I) is directly proportional to Voltage (V) across two points ($V = IR$).
- **Resistance (R):** The property of a conductor that opposes current flow, transforming electrical energy into heat.
- **Resistivity Laws:** Resistance depends on material nature (ρ), length (l), cross-sectional area (A), and temperature.
  - **Formula:** $R = \rho \cdot (l/A)$.
  - **Unit:** Ohm-metre (Ω-m).
- **Power Absorption:** Calculated as $P = VI$, $P = I^2R$, or $P = V^2/R$.

PHASE 2: NETWORK THEOREMS & ANALYSIS

1. Kirchhoff's Laws

- **KCL (Current Law):** The algebraic sum of currents meeting at a junction is zero ($\sum I = 0$). **Total Current In = Total Current Out**.
- **KVL (Voltage Law):** In any closed loop, the algebraic sum of voltage drops (IR) and emfs (E) is zero ($\sum IR + \sum E = 0$).

2. Advanced Simplification Theorems

- **Superposition Theorem:** In a linear circuit with multiple sources, the total current is the algebraic sum of currents produced by each source acting alone, with other sources replaced by their internal resistances.
  - **CBT Trap:** Superposition **cannot** be used to calculate Power directly because power is a non-linear property ($P = I^2R$).
- **Thevenin's Theorem:** Simplifies any linear D.C. network into a single voltage source (V_{th}) in series with a single resistance (R_{th}).
- **Norton's Theorem:** Simplifies a network into a single current source in parallel with a resistance.

PHASE 3: SEMICONDUCTOR PHYSICS & TRANSISTORS

1. Material Properties

- **Intrinsic:** A pure semiconductor (e.g., Silicon or Germanium).
  - **Energy Gaps:** Germanium = **0.72 eV**; Silicon = **1.1 eV**.
- **Extrinsic:** Pure material doped with impurities to increase conductivity.
- **N-Type:** Doped with pentavalent atoms (e.g., Phosphorus, Antimony); **Electrons** are majority carriers.
- **P-Type:** Doped with trivalent atoms (e.g., Boron, Gallium); **Holes** are majority carriers.

2. Bipolar Junction Transistors (BJT)

- A current-controlled switch with three terminals: **Emitter, Base, and Collector**.
  - **Emitter:** Heavily doped to supply majority carriers.
  - **Base:** Very thin and lightly doped middle section.
  - **Collector:** Collects carriers; larger in size than the emitter.

PHASE 4: INTEGRATED CIRCUITS (IC) & FABRICATION

1. Classification & Scales

- **SSI (Small Scale):** < 12 circuits.
- **MSI (Medium Scale):** 12–99 circuits.
- **LSI (Large Scale):** 100–9,999 circuits.
- **VLSI (Very Large):** 10,000–99,999 circuits.
- **ULSI (Ultra Large):** 100,000–999,999 circuits.
- **GSI (Giga Scale):** 1,000,000+ circuits.

2. Fabrication Terminology

- **Epitaxy:** Controlled growth of a crystalline layer on a substrate.
- **Diffusion:** Introducing impurities into selected regions of a wafer.
- **Etching:** Selective removal of regions of semiconductor or metal.
- **Bonding:** Attaching the die to the ceramic substrate and connecting leads.

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PHASE 5: DIGITAL LOGIC & BOOLEAN ALGEBRA

1. Logic Operations & Gates

- **AND Gate:** Output is 1 only if all inputs are 1.
- **OR Gate:** Output is 1 if one or more inputs are 1.
- **NOT Gate:** Performs negation (inverts the signal).
- **NAND/NOR:** Universal Gates; can implement any Boolean function.
- **XOR Gate:** Output is HIGH only when inputs are different.

2. Boolean Theorems & De Morgan's

- **Single-Variable Laws:** $x \cdot 0 = 0$, $x + 1 = 1$, $x + \overline{x} = 1$, and $x \cdot \overline{x} = 0$.
- **De Morgan's Theorems:**
  1. $\overline{A + B} = \overline{A} \cdot \overline{B}$.
  2. $\overline{A \cdot B} = \overline{A} + \overline{B}$.
  - **Mnemonic:** "Break the bar, change the sign!"

3. Combinational Design (The 5-Step SOP Protocol)

1. **Truth Table:** List all 2^N input combinations.
2. **AND Terms:** Write an AND term for every case where output is 1.
3. **SOP Form:** OR these terms together.
4. **Simplify:** Use Boolean algebra or K-maps.
5. **Implement:** Build the circuit using logic gates.

PHASE 6: NUMBER SYSTEMS & BINARY ARITHMETIC

Based on LASU Course Outline and Past Papers

1. The Four Bases of Digital Logic

- **Binary (Base-2):** Digits 0–1. The language of logic gates.
- **Octal (Base-8):** Digits 0–7.
- **Decimal (Base-10):** Digits 0–9. Our everyday system.
- **Hexadecimal (Base-16):** Digits 0–9, A–F (where A=10... F=15).

2. Rapid Conversion & Arithmetic (MCQ Shortcuts)

- **Hex to Binary:** Replace each hex digit with its 4-bit binary equivalent (e.g., $F = 1111$).
- **Binary to Decimal:** Use the weight of positions ($2^0, 2^1, 2^2$...).
- **LASU Past Paper Challenge:** * $BAD_{16} + 3AC_{16}$**:** Perform hex addition column by column.
  - 127.24_8 **to Decimal:** Multiply each digit by 8^n ($1 \times 64 + 2 \times 8 + 7 \times 1 + 2 \times 8^{-1}$...).

PHASE 7: ADVANCED TRANSISTOR NOTATIONS (FETs)

As explicitly asked in LASU 2023 Q1

1. Field Effect Transistor (FET) Notations

Unlike the BJT (Current-controlled), the FET is a **voltage-controlled** device.

- **Source:** Terminal where charge carriers enter the channel.
- **Drain:** Terminal where charge carriers leave the channel.
- **Gate:** Terminal that controls the channel's conductivity.
- **Channel:** The path through which carriers flow between source and drain.

2. Advantages of FETs over BJTs

- Very high input impedance (draws almost no current from the source).
- Less noise and better thermal stability.
- Easier to fabricate in high density on IC chips.

PHASE 8: INTEGRATED CIRCUIT FABRICATION (DEEP DIVE)

Based on LASU IC Terminologies

1. Structuring the Silicon

- **Monolithic IC:** All components (active and passive) are fabricated on a single continuous piece of silicon called a **wafer** or **substrate**.
- **Thick and Thin Film ICs:** Constructed by depositing conductive films (0.1 to $0.5\mu m$) through a mask onto a glass or ceramic substrate.
- **Hybrid (Multichip) ICs:** A combination of film and monolithic techniques.

2. Fabrication Glossary for MCQs

- **Chip:** An extremely small part of a silicon wafer on which the circuit is fabricated.
- **Circuit Probing:** Testing the electrical performance of each IC chip using a microscope and multi-point probe.
- **Metallization:** Depositing aluminum over the wafer to provide ohmic contacts and interconnections.
- **Photolithography:** Transferring a geometrical pattern from a mask to the wafer surface.

PHASE 9: COMBINATIONAL LOGIC & SIMPLIFICATION TOOLS

Based on Boolean Mastery and K-Maps

1. Karnaugh Maps (K-Maps)

- A graphical tool for minimizing Boolean expressions.
- **Gray Code Adjacency:** Only one variable changes between adjacent cells.
- **Grouping Rules:** Groups must be powers of 2 ($2, 4, 8, 16$). Larger groups eliminate more variables.

2. Sum-of-Products (SOP) vs. Product-of-Sums (POS)

- **SOP:** Terms like $AB + CD$. Preferred for NAND-NAND implementation.
- **POS:** Terms like $(A + B)(C + D)$. Preferred for NOR-NOR implementation.

PHASE 10: ANALOG/DIGITAL INTERFACING

Based on ADC/DAC Principles

1. The Digital Advantage

- **Programming:** Operation is easily controlled by stored instructions.
- **Accuracy:** Digitized signals do not deteriorate with processing, whereas analog signals distort due to temperature and component tolerance.
- **Density:** More digital circuitry can be fabricated on IC chips because they avoid bulky components like high-value capacitors or inductors.

PHASE 11: EXCLUSIVE LOGIC GATES (XOR/XNOR)

Essential for parity checkers and arithmetic circuits.

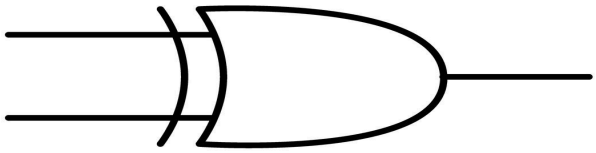
1. Exclusive-OR (XOR) Gate

- **Logic Rule**: The output is HIGH (1) only when the inputs are **different**.
- **Boolean Expression**: $x = A \oplus B$ (Standard) or $x = \overline{A}B + A\overline{B}$ (Expansion).
- **Truth Table Summary**:
  - $0 \oplus 0 = 0$
  - $0 \oplus 1 = 1$
  - $1 \oplus 0 = 1$
  - $1 \oplus 1 = 0$

2. Exclusive-NOR (XNOR) Gate

- **Logic Rule**: The output is HIGH (1) only when the inputs are the **same**.
- **Boolean Expression**: $x = \overline{A \oplus B} = AB + \overline{AB}$.
- **Application**: Used as a digital comparator to determine if two binary bits are identical.

XOR



Gate	Notation	Truth table
XOR	$A \oplus B$	

INPUT		OUTPUT
A	B	A XOR B
0	0	0
0	1	1
1	0	1
1	1	0

PHASE 12: MULTIVIBRATORS & SEQUENTIAL BASICS

As highlighted in your 2023/2024 Exam Question 3(e).

Unlike combinational circuits, sequential circuits (like multivibrators) often have **memory**.

- **Astable Multivibrator**: Has no stable state; it continually switches between HIGH and LOW (Square-wave generator/Oscillator).
- **Monostable Multivibrator**: Has one stable state. When triggered, it stays in an unstable state for a set time before returning (Timer/One-shot).
- **Bistable Multivibrator**: Has two stable states. It remains in one state until triggered into the other (The basic Flip-Flop).

PHASE 13: ADVANCED IC STRUCTURES & PACKAGING

Detailed classification based on structural fabrication.

1. Structural Classifications

- **Monolithic ICs**: All components (active and passive) are fabricated indissolubly on a single continuous piece of silicon (wafer).
- **Thick/Thin Film ICs**: Constructed by depositing conductive films (0.1 to 0.5 μm) through a mask onto a glass or ceramic substrate.
- **Hybrid (Multichip) ICs**: Formed by interconnecting a number of individual chips or by a combination of film and monolithic techniques.

2. Critical Terminology

- **Die/Chip**: An extremely small part of a silicon wafer on which the circuit is fabricated.
- **Bonding**: The process of attaching the die to the ceramic substrate and connecting the leads to the package.
- **Encapsulation**: Putting a cap over the IC and sealing it in an inert atmosphere to protect it from the environment.

PHASE 14: THE MASTER CALCULATION REFERENCE SHEET

Step-by-step logic for the most frequent calculation types in LASU CSC 217 papers.

1. The Coil Resistance Problem

- **Scenario**: 2000 turns, $0.8\ \mathrm{mm}^2$ area, $80\ \mathrm{cm}$ per turn, resistivity $\rho = 0.02\ \mu\Omega\text{-m}$.
- **Step 1: Total Length (L)**: $2000 \times 80\ \mathrm{cm} = 160000\ \mathrm{cm} = 1600\ \mathrm{m}$.
- **Step 2: Standard Area (A)**: $0.8\ \mathrm{mm}^2 = 0.8 \times 10^{-6}\ \mathrm{m}^2$.
- **Step 3: Resistivity (ρ)**: $0.02 \times 10^{-6}\ \Omega\text{-m}$.
- **Step 4: Solve R**: $R = (0.02 \times 10^{-6} \times 1600)/(0.8 \times 10^{-6}) = 40\ \Omega$.
- **Step 5: Power (P)**: $P = V^2/R = 110^2/40 = 302.5\ \mathrm{W}$.

2.

Binary-Value Comparator Logic

- **Goal**: Design a circuit that is HIGH only if a 4-bit binary number ($ABCD$) is > 0110 (decimal 6).
- **Truth Table Rows**: $2^4 = 16$ rows.
- **Active States**: Output is 1 for binary values 7 through 15.

PHASE 15: SEMICONDUCTOR PHYSICS (BEYOND THE BASICS)

Deepening your understanding of the physical behavior of Silicon and Germanium.

1. Energy Band Theory

- **Valence Band:** The lower energy band that is almost completely filled with electrons.
- **Conduction Band:** The upper energy band that is almost empty.
- **Forbidden Energy Gap (E_g):** The narrow gap separating the valence and conduction bands.
  - Electrons must possess enough energy to "jump" across this gap to become available for conduction.
  - **Silicon Gap:** $1.1\ \mathrm{eV}$.
  - **Germanium Gap:** $0.72\ \mathrm{eV}$.

2. Carrier Dynamics: Electrons and Holes

- **Hole Formation:** When an electron jumps into the conduction band, it leaves behind a positively charged vacancy in the valence band called a **hole**.
- **Total Current:** In a semiconductor, the total current is the sum of electron current and hole current flowing in opposite directions.
- **Fermi Level (E_F):** * In an **Intrinsic** semiconductor, the Fermi level lies exactly in the middle of the forbidden energy gap.
  - In **N-Type**, the Fermi level shifts upward toward the conduction band.
  - In **P-Type**, the Fermi level shifts downward toward the valence band.

PHASE 16: BUBBLED LOGIC & GATE EQUIVALENCIES

Essential for "visual" simplification in CBT exams.

1. The "Bubbled" Equivalents (De Morgan in Circuits)

- **NOR Gate:** Equivalent to an **AND gate with inverted inputs** (Bubbled-AND).
- **NAND Gate:** Equivalent to an **OR gate with inverted inputs** (Bubbled-OR).

2. Universality in Action

- **NAND as INVERTER:** Connect both inputs of a NAND gate together.
- **NAND as AND:** Follow a NAND gate with a NAND-inverter.
- **NAND as OR:** Use two NAND-inverters at the inputs of a third NAND gate.

PHASE 17: IC FABRICATION & PACKAGING TERMINOLOGY

Detailed steps required for the "Vocabulary" portion of digital electronics.

1. **Substrate Preparation:** Starting with a thin disk of single-crystal silicon called a **wafer**.
2. **Epitaxy:** Growing a thin, controlled layer of single-crystal silicon on the wafer substrate.
3. **Photolithography:** Using light and a mask to transfer a specific geometric pattern to the wafer surface.
4. **Diffusion:** Introducing impurities (doping) into selected regions of the wafer to form P-N junctions.
5. **Etching:** Selectively removing regions of the semiconductor or metal using chemical processes.
6. **Metallization:** Evaporating aluminum over the wafer to provide ohmic contacts and circuit interconnections.
7. **Circuit Probing:** Testing the electrical performance of each chip while still on the wafer using a microscope and probes.
8. **Encapsulation:** Sealing the chip in a protective cap (inert atmosphere) to ensure long-term reliability.

PHASE 18: DESIGN WORKSHOP (EXAM SCENARIOS)

Walkthroughs of common LASU logic design questions.

1. The Spaceship Battery Monitor (Example 4-8)

- **Requirement:** Produce a HIGH output if battery voltage $> 0110_2$ (decimal 6).
- **Logic:** The output x is 1 for binary states 7, 8, 9, 10, 11, 12, 13, 14, and 15.
- **SOP Simplification:** Resulting simplified expression usually centers on the MSB (A) and other high-value bits ($A + BC + BD$).

2. The Majority Logic Circuit

- **Requirement:** Output is HIGH only when a majority of 3 inputs (A, B, C) are HIGH.
- **Truth Table:** Output is 1 for cases $011, 101, 110$, and 111.
- **Simplified Expression:** $x = AB + BC + AC$.
  - *Note: This requires **three** 2-input AND gates and **one** 3-input OR gate.*

PHASE 19: THE BINARY & HEXADECIMAL BATTLEFIELD

In a CBT exam, conversion speed is your greatest weapon.

1. Rapid Base Conversions

- **Hex to Binary:** Replace each Hex digit with a 4-bit binary string (e.g., $C_{16} = 1100_2$).
- **Octal to Binary:** Replace each Octal digit with a 3-bit binary string (e.g., $7_8 = 111_2$).
- **The Weighting Rule:** For binary-to-decimal, use the powers of 2 $(1, 2, 4, 8, 16, 32...)$.
- **Decimal to Binary:** Use the repeated division-by-2 method; the remainder gives you the bits from **LSB to MSB**.

2. Hex Arithmetic (LASU Favorite)

- When adding Hex numbers (like $BAD + 3AC$):
  - $D(13) + C(12) = 25.\ 25 - 16 = 9$ (carry 1).
  - $A(10) + A(10) + 1 = 21.\ 21 - 16 = 5$ (carry 1).
  - $B(11) + 3 + 1 = 15 = F$.
  - Result: $F59_{16}$.

PHASE 20: MULTIVIBRATORS & SEQUENTIAL LOGIC

Based on LASU Exam Question 3(e)

Unlike combinational circuits, sequential circuits have **memory** and often use multivibrators to generate clock pulses.

- **Astable Multivibrator:** Has **no stable state**. It continuously toggles between HIGH and LOW.
  - **Application:** Square-wave or clock-pulse generator.
- **Monostable Multivibrator:** Has **one stable state**. It returns to its stable state after a set time.
  - **Application:** Digital timers or "one-shot" pulse generators.
- **Bistable Multivibrator:** Has **two stable states**. It stays in one state until a trigger forces it to the other.
  - **Application:** This is the foundation of the **Flip-Flop** (the basic memory unit).

PHASE 21: THE IC FABRICATION ENCYCLOPEDIA

Terms frequently tested in the "Vocabulary" portion of CSC 217

- **Substrate/Wafer:** The thin silicon disk that serves as the foundation for the entire IC.
- **Epitaxy:** Growing a controlled crystalline layer on the substrate.
- **Diffusion:** Introducing dopants (P-type or N-type) into specific regions to create junctions.
- **Etching:** Selectively removing regions of metal or semiconductor.
- **Photolithography:** Using light and a mask to transfer geometric patterns to the chip.
- **Metallization:** Evaporating aluminum over the chip to create interconnections and ohmic contacts.
- **Circuit Probing:** Testing individual chips while still on the wafer using a multi-point probe.
- **Encapsulation:** Sealing the final chip in a protective cap with an inert atmosphere.

PHASE 22: THE RESISTIVITY CALCULATION "GOD MODE"

Step-by-step logic for the most repeated calculation in your notes

Problem Context: 2000 turns, $0.8\ \mathrm{mm}^2$ area, $80\ \mathrm{cm}$ per turn, $\rho = 0.02\ \mu\Omega\text{-m}$.

1. **Standardize Units (The MCQ Survival Step):**
   - $L = 2000 \times 0.8\ \mathrm{m} = 1600\ \mathrm{m}$.
   - $A = 0.8 \times 10^{-6}\ \mathrm{m}^2$ (Critical conversion!).
   - $\rho = 0.02 \times 10^{-6}\ \Omega\text{-m}$.
2. **Calculate Resistance (R):**
   - $R = (0.02 \times 10^{-6} \times 1600)/(0.8 \times 10^{-6}) = \mathbf{40\ \Omega}$.
3. **Calculate Power (P):**
   - If $V = 110\ \mathrm{V}$, $P = V^2/R = 12100/40 = \mathbf{302.5}\ \mathrm{W}$.

THE BOOLEAN THEOREM DIRECTORY

1. Single-Variable Rules

- **AND Operations:**
  - $x \cdot 0 = 0$
  - $x \cdot 1 = x$
  - $x \cdot x = x$
  - $x \cdot \overline{x} = 0$
- **OR Operations:**
  - $x + 0 = x$
  - $x + 1 = 1$
  - $x + x = x$
  - $x + \overline{x} = 1$

2. Multivariable "Quick-Hits"

- **Commutative Law:** $x + y = y + x$ and $xy = yx$.
- **Associative Law:** $x + (y + z) = (x + y) + z$.
- **Distributive Law:** $x(y + z) = xy + xz$.
- **Absorptive Rule:** $x + xy = x$.
- **Elimination Rule:** $x + \overline{x}y = x + y$.

DE MORGAN'S THEOREMS & GATE EQUIVALENCIES

1. The Two Pillars

1. **Theorem 1:** The complement of a sum is equal to the product of the complements: $\overline{x+y} = \overline{x} \cdot \overline{y}$.
2. **Theorem 2:** The complement of a product is equal to the sum of the complements: $\overline{x \cdot y} = \overline{x} + \overline{y}$.

2. Circuit Implications

- **Bubbled-AND:** An AND gate with inverters on all inputs is functionally a **NOR gate**.
- **Bubbled-OR:** An OR gate with inverters on all inputs is functionally a **NAND gate**.

IC FABRICATION & TERMINOLOGY

A master glossary for "Vocabulary" questions.

- **Wafer:** The thin silicon disk used for simultaneous fabrication of many ICs.
- **Epitaxy:** The controlled growth of a crystalline layer on a substrate.
- **Photolithography:** Transferring patterns from a mask to the wafer using ultraviolet light.
- **Diffusion:** Introducing dopant impurities into specific regions to create junctions.
- **Etching:** Selectively removing regions of the semiconductor or metal surface.
- **Metallization:** Evaporating aluminum over the wafer for interconnections.
- **Encapsulation:** Sealing the IC in a protective cap within an inert atmosphere.

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CSC 217 DIGITAL ELECTRONICS PRACTICE EXAM QUESTIONS

1. **What is the internal resistance of an ideal voltage source?** A. Infinite B. Zero C. $1\ \Omega$ D. $100\ \mathrm{k}\Omega$
2. **An ideal current source provides a constant current regardless of:** A. The internal resistance B. The supply frequency C. The voltage across its terminals D. The direction of flow
3. **Kirchhoff's Current Law (KCL) states that the algebraic sum of currents meeting at a junction is:** A. One B. Positive C. Zero D. Negative
4. **KCL is based on the principle of conservation of:** A. Energy B. Charge C. Mass D. Momentum
5. **Which theorem allows a complex linear D.C. network to be replaced by a single voltage source and a series resistor?** A. Superposition Theorem B. Norton's Theorem C. Thevenin's Theorem D. Ohm's Law
6. **In Superposition Theorem, when considering one voltage source, all other ideal voltage sources are replaced by:** A. Open circuits B. Short circuits C. Resistors D. Diodes
7. **Kirchhoff's Voltage Law (KVL) is primarily based on the conservation of:** A. Charge B. Power C. Energy D. Voltage
8. **Superposition Theorem cannot be used to calculate which of the following directly?** A. Voltage B. Current C. Power D. Resistance
9. **The property of a conductor that opposes the passage of current is:** A. Capacitance B. Inductance C. Resistance D. Conductance
10. **The S.I. unit of resistivity is:** A. Ohms (Ω) B. Ohm-metre (Ω-m) C. Amperes (A) D. Volts (V)

11. **What is the energy gap of Silicon at room temperature?** A. $0.72\ \mathrm{eV}$ B. $1.1\ \mathrm{eV}$ C. $2.0\ \mathrm{eV}$ D. $0.5\ \mathrm{eV}$
12. **In an N-type semiconductor, the majority charge carriers are:** A. Holes B. Neutrons C. Electrons D. Protons
13. **Which impurity is added to pure Silicon to create a P-type semiconductor?** A. Phosphorus B. Antimony ...**Boron** (Trivalent) D. Arsenic
14. **A pentavalent impurity atom is also known as a:** A. Acceptor atom B. Neutral atom C. Donor atom D. Hole creator
15. **The Fermi level in an intrinsic semiconductor lies:** A. Near the conduction band B. Near the valence band C. Exactly in the middle of the energy gap D. Inside the conduction band
16. **Which BJT terminal is heavily doped to supply majority carriers?** A. Base B. Collector C. Emitter D. Gate
17. **A Bipolar Junction Transistor (BJT) acts primarily as a:** A. Voltage-controlled switch B. Current-controlled switch C. Resistance-controlled switch D. Power-controlled switch
18. **The base of a transistor is characterized by being:** A. Thick and heavily doped B. Thin and heavily doped C. Thin and lightly doped D. Thick and lightly doped
19. **What is the energy gap for Germanium?** A. $1.1\ \mathrm{eV}$ B. $0.3\ \mathrm{eV}$ C. $0.72\ \mathrm{eV}$ D. $1.5\ \mathrm{eV}$
20. **In a semiconductor, total current is the sum of:** A. Electron and neutron flow B. Electron and hole flow C. Proton and hole flow D. Voltage and Resistance

21. **A logic circuit with 4 inputs will have how many rows in its truth table?** A. 8 B. 12 C. 16 D. 32
22. **Which gate output is HIGH (1) only if all its inputs are HIGH?** A. OR Gate B. NAND Gate C. AND Gate D. XOR Gate
23. **The output of an OR gate is LOW (0) only when:** A. Any input is 1 B. All inputs are 1 C. All inputs are 0 D. One input is 0 and the other is 1
24. **Which gates are referred to as "Universal Gates"?** A. AND & OR B. NAND & NOR C. XOR & XNOR D. NOT & AND
25. **The XOR gate produces a HIGH output only when its inputs are:** A. The same B. Both LOW C. Different D. Both HIGH
26. **A NOT gate is also known as a/an:** A. Buffer B. Inverter C. Summer D. Multiplier
27. **The Boolean expression for an OR operation is denoted by:** A. $A \cdot B$ B. $A \oplus B$ C. $A + B$ D. $\overline{A}$
28. **An AND gate with inverted inputs is equivalent to which gate?** A. NAND B. NOR C. XOR D. XNOR
29. **Which gate performs the "Sum-of-Products" final operation?** A. AND B. OR C. NOT D. NAND
30. **To use a NAND gate as an inverter, you must:** A. Ground one input B. Connect the inputs together C. Connect one input to V_{cc} D. Leave inputs floating

31. **De Morgan's first theorem states that $\overline{A+B}$ is equal to:** ...$\overline{A} \cdot \overline{B}$ B. $\overline{A} + \overline{B}$ C. $A \cdot B$ D. $\overline{A \cdot B}$
32. **According to Boolean algebra, $X + \overline{X}$ equals:** A. 0 B. X C. 1 D. $2X$
33. **Simplifying the expression $X + XY$ results in:** A. Y B. XY C. X D. 1
34. **The term "SOP" in digital electronics stands for:** A. Sum of Products B. System of Parts C. Standard Operating Procedure D. Series of Power
35. **Karnaugh Maps (K-maps) are used to:** A. Design analog circuits B. Minimize Boolean expressions C. Calculate power dissipation D. Increase circuit complexity
36. **In a K-map, groups must always be in powers of:** A. 3 B. 10 C. 2 $(1, 2, 4, 8...)$ D. 5
37. **The expression $\overline{A \cdot B}$ is equivalent to:** A. $\overline{A} \cdot \overline{B}$ B. $\overline{A} + \overline{B}$ C. $A + B$ D. $\overline{A+B}$
38. **The "Majority Logic" expression for a 3-input circuit is:** A. $A + B + C$ B. ABC C. $AB + BC + AC$ D. $\overline{ABC}$
39. **In Boolean algebra, $X \cdot 0$ equals:** A. X B. 1 C. 0 D. $\overline{X}$
40. **Distributive Law states that $A(B + C)$ equals:** A. $A + BC$ B. $AB + AC$ C. ABC D. $(A + B)(A + C)$

41. **SSI (Small Scale Integration) typically contains how many gates per chip?** A. More than 100 B. Less than 12 C. Exactly 50 D. 1000
42. **Which of the following cannot be easily fabricated inside an IC?** A. Resistors B. Transistors C. Inductors (Coils) D. Diodes
43. **Very Large Scale Integration (VLSI) involves how many circuits per package?** A. $10 - 99$ B. $100 - 999$ C. $10,000 - 99,999$ D. < 10
44. **What is the primary advantage of Integrated Circuits over discrete circuits?** A. Higher power handling B. Ease of repair C. Small size and high reliability D. Ability to fabricate large coils
45. **The process of introducing impurities into selected regions of a silicon wafer is:** A. Epitaxy B. Etching C. Diffusion D. Metallization
46. **What is a "Monolithic IC"?** A. A circuit built with vacuum tubes B. A circuit where all components are fabricated on a single silicon chip C. A circuit made of only resistors D. A circuit that uses discrete transistors soldered together
47. **The process of depositing aluminum to create interconnections on an IC is:** A. Photolithography B. Encapsulation C. Metallization D. Probing
48. **Linear Integrated Circuits (LICs) are also known as:** A. Digital ICs B. Power ICs C. Analog ICs D. Binary ICs
49. **What is the main reason digital systems are easier to design than analog ones?** A. They use higher voltage B. Exact voltage values aren't critical, only the range (HIGH/LOW) C. They don't use transistors D. They are always mechanical
50. **Which device converts real-world physical quantities into digital form?** A. DAC B. Transistor C. ADC (Analog-to-Digital Converter) D. Transformer

CSC 217 DIGITAL ELECTRONICS PRACTICE EXAM QUESTIONS ANSWERS.

1. B | 2. C | 3. C | 4. B | 5. C | 6. B | 7. C | 8. C | 9. C | 10. B

11. B | 12. C | 13. C | 14. C | 15. C | 16. C | 17. B | 18. C | 19. C | 20. B

21. C | 22. C | 23. C | 24. B | 25. C | 26. B | 27. C | 28. B | 29. B | 30. B

31. A | 32. C | 33. C | 34. A | 35. B | 36. C | 37. B | 38. C | 39. C | 40. B

41. B | 42. C | 43. C | 44. C | 45. C | 46. B | 47. C | 48. C | 49. B | 50. C

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